

Histogram

B_j : j th bin width b_j

$$\hat{p}(x) = \frac{1}{nb_j} \sum_{i=1}^n \mathbb{I}(X_i \in B_j), \quad x \in B_j$$

$$\mathcal{F} = \{p: p \geq 0, \int p = 1, p = c_j \text{ on } B_j, c_j \in \mathbb{R}^+\}$$

$$* f_j = \sum_{i=1}^n \mathbb{I}(X \in B_j) \text{ bin frequency}$$

$$\text{then } \hat{p} = \arg \max_{p \in \mathcal{F}} \prod_{i=1}^n p(X_i)$$

$$\begin{aligned} \text{p.f. } L(f) &= \sum_{i=1}^n \log f(X_i) = \sum_{i=1}^n \log \left\{ \sum_{j=1}^m c_j \mathbb{I}(X_i \in B_j) \right\} \\ &= \sum_{j=1}^m \left\{ \sum_{i=1}^n \mathbb{I}(X_i \in B_j) \right\} \log c_j \\ &= \sum_{j=1}^m f_j \mathbb{I}(f_j > 0) \log c_j \end{aligned}$$

$$\begin{aligned} &\text{maximize } L(c_1, \dots, c_m) \\ &\text{subject to } \sum_{j=1}^m b_j c_j = 1. \end{aligned}$$

Let $f_{(1)}, \dots, f_{(k)} > 0, a_j = b_j c_j$ then

$$L(a_1, \dots, a_k) = L(a_{(1)}, \dots, a_{(k)}) = \sum_{i=1}^k f_{(i)} \log a_{(i)}$$

$$= \sum_{i=1}^{k-1} f_{(i)} \log a_{(i)} + f_{(k)} \log \left\{ 1 - \sum_{i=1}^{k-1} a_{(i)} \right\}$$

$$\frac{\partial L}{\partial a_{(i)}} = \frac{f_{(i)}}{a_{(i)}} - \frac{f_{(k)}}{a_{(k)}}$$

$\frac{\partial^2 L}{\partial a_{(i)} \partial a_{(j)}} = \text{negative definite}$

$$f_{(i)}/a_{(i)} = \text{const.} = \frac{n}{I}$$

$$\hat{a}_{(i)} = \frac{f_{(i)}}{n}, \quad \hat{c}_{(i)} = \frac{f_{(i)}}{nb_{(i)}} \quad \square$$

Assume $b_j \equiv h, B_j = [(j-1)h, jh)$.

Thm. As $h \rightarrow 0, nh \rightarrow \infty, p$ is continuously differentiable on a nhbd of x .

$$E(\hat{p}(x)) - p(x) = \left\{ \underbrace{(j(x) - \frac{1}{2})h - x}_{\text{index } x \text{ belongs.}} \right\} p'(x) + o(h).$$

$$\text{var}(\hat{p}(x)) = \frac{1}{nh} p(x) + o\left(\frac{1}{nh}\right)$$

pA $f_j \sim B(n, \mathbb{P}(X_i \in B_j))$.

$$E(\hat{p}(x)) = E\left(\frac{1}{nh} f_j\right) = \frac{1}{n} \mathbb{P}(X_i \in B_j) = \frac{1}{n} \int_{(j-1)h}^{jh} p(t) dt$$

$$\begin{aligned} E(\hat{p}(x)) - p(x) &= \frac{1}{n} \int_{(j-1)h}^{jh} \{p(t) - p(x)\} dt \leftarrow \int_{(j-1)h}^{jh} p(t) - p(x) = (t-x)p'(x) + \frac{(t-x)^2}{2} p''(x) + \dots \\ &= \left\{ (j - \frac{1}{2})h - x \right\} p'(x) + \int_{(j-1)h}^{jh} \frac{(t-x)}{h} \cdot \{p'(a+u(t-x)) - p'(x)\} dt du = II \end{aligned}$$

$$|II| \leq \int_0^1 \int_{(j-1)h}^{jh} \left| \frac{t-x}{h} \right| |f'(a+u(t-x)) - f'(x)| dt du$$

$$\leq \int_0^1 \int_{(j-1)h}^{jh} 1 \cdot \sup_{|y| \leq h} |f'(a+uy) - f'(x)| dt du = o(h).$$

#.

$$\text{var}(\hat{f}_h(x)) = \frac{\text{var}(f_j)}{n^2 h^2} = \frac{1}{nh^2} \underbrace{P(X \in B_j)}_{\rightarrow 0(1)} (1 - \underbrace{P(X \in B_j)}_{\rightarrow 0(1)})$$

$$= \frac{1}{nh} \left(\frac{1}{h} \int_{j-1}^j p(x) dx \right) (1 - o(1))$$

$$= \frac{1}{nh} (p(x) + o(1)) (1 - o(1)) = \frac{1}{nh} p(x) + o\left(\frac{1}{nh}\right).$$

□

L^2 -error.

Thm. As $h \rightarrow 0$, $nh \rightarrow \infty$ and p bounded, continuously differentiable. $\int (p')^2 < \infty$.

$$\int (\mathbb{E} \hat{p}_h(x) - p(x))^2 dx \leq \frac{1}{12} h^2 \int (p')^2 + o(h^2).$$

$$\int \text{var}(\hat{p}_h(x)) dx = \frac{1}{nh} + o\left(\frac{1}{nh}\right).$$

$$\text{MISE}(\hat{p}) = \frac{1}{nh} + \frac{1}{12} h^2 \int (p')^2 + o\left(\frac{1}{nh} + h^2\right).$$

$$\text{pf} \int \text{bias}^2 dx = \sum_k \int_{B_k} \text{bias}^2 dx$$

$$= \sum_k \int_{B_k} \left((k - \frac{1}{2})h - x \right)^2 f' \left((k - \frac{1}{2})h \right)^2 dx + o(h^2).$$

(Here $\text{bias} = (j - \frac{1}{2})h - x$) $f'((j - \frac{1}{2})h)$ ~~$\pm h \cdot \sup_{|y-x| \leq h} |f'(x+y) - f'(x)|$~~ $\pm h \cdot \sup_{|y-x| \leq h} |f'(x+y) - f'(x)|$ $\hookrightarrow$ Uniform on the bin)

$$= \sum_k f' \left((k - \frac{1}{2})h \right)^2 \int_{B_k} \left(x - (k - \frac{1}{2})h \right)^2 dx + o(h^2)$$

$$\approx \frac{1}{3} \cdot \left(\frac{h}{2} \right)^3$$

$$= \frac{1}{12} h^2 \sum_k f' \left((k - \frac{1}{2})h \right)^2 + o(h^2) = \frac{1}{12} h^2 \int f'^2 + o(h^2)$$

$$= \int f'^2 + o(1)$$

$$\int \text{var.} = \sum_k \int_{B_k} \frac{1}{nh^2} P(X \in B_j) (1 - P(X \in B_j)) dx$$

$$= \frac{1}{nh} \sum_k P(X \in B_j) (1 - P(X \in B_j)) = \frac{1}{nh} - \frac{1}{nh} \left(\sum_k P(X \in B_j)^2 \right) = \frac{1}{nh} + o\left(\frac{1}{nh}\right)$$

□

Average Shifted Histogram.

$$\hat{p}_{m,I}(x) = \frac{1}{nh} \sum_{i=1}^m \sum_{j \in \mathbb{Z}} I_{A_{j,I}}(x) I_{A_{j,I}}(x_i)$$

$$A_{j,I} = \left[(j-1)h + I \frac{h}{m}, jh + I \frac{h}{m} \right).$$

$\hookrightarrow$ bin by shifting.

$$\hat{p}_m(x) = \frac{1}{m} \sum_{i=1}^m \hat{p}_{m,I}(x_i).$$

Thm. Limit of ASH.

$$\begin{aligned}
 \hat{f}(x) &= \frac{1}{nh} \sum_{i=1}^n \sum_{\ell=-m}^{m-1} \int I_{A_{j,\ell}}(x) I_{A_{j,\ell}}(x_i) dx \\
 &= \frac{1}{nh} \sum_{i=1}^n \sum_{\ell=-m}^{m-1} (m-|\ell|) I_{I_\ell}(x) I_{I_{\ell+1}}(x_i) \\
 &= \frac{1}{nh} \sum_{i=1}^n \sum_{\ell=-m}^{m-1} \left(1 - \frac{|\ell|}{m}\right) I_{I_\ell}(x) I_{I_{\ell+1}}(x_i) \\
 &= \frac{1}{nh} \sum_{i=1}^n \left(1 - \frac{|j(x) - j(x_i)|}{m}\right) I\left(\frac{|j(x) - j(x_i)|}{m} < 1\right) \\
 &\xrightarrow{m \rightarrow \infty} \frac{1}{nh} \sum_{i=1}^n \left(1 - \left|\frac{x_i - x}{h}\right|\right) I\left(\left|\frac{x_i - x}{h}\right| < 1\right) \\
 &\quad \quad \quad K(x) = (1 - |x|) I_{[-1,1]}(x) \quad K_h(x) = K\left(\frac{x - x_i}{h}\right)
 \end{aligned}$$

Kernel Density Estimator, $\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right) = \frac{1}{n} \sum_{i=1}^n K_h(x - x_i)$

- Assumption (A1). $p^{(r)}$ is continuous at x and $p^{(r)} \in L^1$.
 (A2). $|u|^r K(u) \rightarrow 0$ as $|u| \rightarrow \infty$, $\int |u|^r K(u) du < \infty$.

Theorem 2.1. Under (A1), (A2), as $h \rightarrow 0$.

$$(K_h * f)(x) = \sum_{j=0}^r \frac{(-1)^j \mu_j(K)}{j!} p^{(j)}(x) h^j + o(h^r)$$

where $\mu_j(K) = \int u^j K(u) du$.

pf $f(x-u) = \sum_{j=0}^r \frac{(-1)^j u^j}{j!} f^{(j)}(x) + \frac{(-1)^r u^r}{(r-1)!} \int_0^1 (f^{(r)}(x-wu) - f^{(r)}(x)) \cdot (1-w)^{r-1} dw$

$$\begin{aligned}
 (K_h * f)(x) &= \int \frac{1}{h} K\left(\frac{x-u}{h}\right) f(u) du \\
 &= \int K(t) f(x-hu) dt \\
 &= \sum_{j=0}^r \frac{(-1)^j \mu_j(K)}{j!} \cdot f^{(j)}(x) h^j + R_n(x)
 \end{aligned}$$

$$R_n(x) = \frac{(-1)^r h^r}{(r-1)!} \int_0^1 \int_0^1 t^r K(t) \{f^{(r)}(x-wht) - f^{(r)}(x)\} (1-w)^{r-1} dw dt$$

$$\frac{(r-1)! |R_n(x)|}{h^r (r-1)!} \leq \int_0^1 \int_0^1 |t|^r K(t) |f^{(r)}(x-wht) - f^{(r)}(x)| (1-w)^{r-1} dw dt$$

$$= \int_0^1 \int_{|t| \leq \frac{\delta}{h}} |t|^r K(t) |f^{(r)}(x-wht) - f^{(r)}(x)| (1-w)^{r-1} dt dw$$

$$+ \int_0^1 \int_{|t| > \frac{\delta}{h}} |t|^r K(t) |f^{(r)}(x-wht) - f^{(r)}(x)| (1-w)^{r-1} dt dw$$

$$(First) \leq \int_0^1 \int_{|t| \leq \frac{\delta}{\omega h}} |t|^{r-1} K(t) |f^{(r)}(x-h\omega t) - f^{(r)}(x)| (1-\omega)^{r-1} dt d\omega.$$

$$\leq \sup_{|y| \leq \delta} |f^{(r)}(x-y) - f^{(r)}(x)| \cdot \int |t|^{r-1} K(t) dt.$$

$$(Second) = \int_0^1 \int_{|t| > \frac{\delta}{\omega h}} |t|^{r-1} K(t) |f^{(r)}(x-h\omega t) - f^{(r)}(x)| (1-\omega)^{r-1} dt d\omega.$$

$$\leq \int_0^1 \int_{|t| > \frac{\delta}{\omega h}} |t|^{r-1} K(t) |f^{(r)}(x)| dt d\omega + \int_0^1 \int_{|t| > \frac{\delta}{\omega h}} |t|^{r-1} K(t) |f^{(r)}(x-h\omega t)| dt d\omega.$$

$$\leq \int_{|t| > \frac{\delta}{\omega h}} |t|^{r-1} K(t) |f^{(r)}(x)| dt + \int_0^1 \int_{|t| > \frac{\delta}{\omega h}} \frac{\omega h}{|t|^{r+1}} |t|^{r+1} K(t) |f^{(r)}(x-h\omega t)| dt d\omega.$$

$$\leq \left(\int_{|t| > \frac{\delta}{h}} |t|^{r-1} K(t) dt \right) \cdot |f^{(r)}(x)| + \max_{|t| > \frac{\delta}{h}} \{ |t|^{r+1} K(t) \} \cdot \int |f^{(r)}(y)| dy \cdot \frac{1}{\delta}.$$

$$\frac{(r-1)!}{(h)^{r-1}} |R_h(x)| \leq \max_{|y| \leq \delta} |f^{(r)}(x-y) - f^{(r)}(x)| \int |t|^{r-1} K(t) dt + |f^{(r)}(x)| \cdot \int_{|t| > \frac{\delta}{h}} |t|^{r-1} K(t) dt + \max_{|t| > \frac{\delta}{h}} \{ |t|^{r+1} K(t) \} \cdot \frac{1}{\delta} \int |f^{(r)}(y)| dy.$$

$$\lim_{h \rightarrow 0} \frac{(r-1)!}{(h)^{r-1}} |R_h(x)| \leq \max_{|y| \leq \delta} |f^{(r)}(x-y) - f^{(r)}(x)| \int |t|^{r-1} K(t) dt$$

as $\delta \downarrow 0$.

□

Assumption. A3. p'' is continuous at x , $p'' \in L^1$.

A4. K is symmetric, $|u|^2 K(u) \rightarrow 0$ as $|u| \rightarrow \infty$, $\int |u|^2 K(u) du < \infty$, $\int K = 1$.

A5. p is continuous at x .

A6. $|u| K(u) \rightarrow 0$ as $|u| \rightarrow \infty$, $\int K^2 < \infty$.

Thm. A3, A4.

$$E(\hat{p}(x)) - p(x) = \frac{1}{2} \mu_2(K) p''(x) h^2 + o(h^2).$$

Thm. A5, A6

$$\text{var}(\hat{p}(x)) = \mu_0(K^2) p(x) \cdot \frac{1}{nh} + o\left(\frac{1}{nh}\right).$$

pf. $r=2$; (A1), (A2) holds, K symmetric, $\mu_1(K)=0$

$$\hat{p}(x) = p(x) + \frac{1}{2} \mu_2(K) p''(x) h^2 + o(h^2)$$

$r=0$; (A5), (A6) holds,

$$\hat{p}(x) = \frac{1}{n} \sum_{i=1}^n K_h(x - X_i). \quad \text{var}(\hat{p}(x)) = \frac{1}{n} \text{var}(K_h(x - X_1)).$$

$$\begin{aligned} \text{Var}(K_h(x-X_1)) &= \int \frac{1}{h} k\left(\frac{x-x_1}{h}\right)^2 p(x) dx - \left(\int \frac{1}{h} k\left(\frac{x-x_1}{h}\right) p(x) dx \right)^2 \\ &= \frac{1}{h} (K_h^2 * p)(x_1) - (K_h * p(x_1))^2 \\ &= \frac{1}{h} (\mu_0(K^2) \cdot p(x_1) + o(1)) - (p(x_1) + o(1))^2 \\ &= \frac{1}{h} \mu_0(K^2) p(x_1) + o\left(\frac{1}{h}\right) \end{aligned}$$

$$\text{var}(\hat{p}(x)) = \frac{1}{nh} \mu_0(K^2) p(x_1) + o\left(\frac{1}{nh}\right).$$

L^2 error.
Thm 2.2 p'' continuous, $\int (p'')^2 < \infty$. Assume k is symmetric, $\int |u|^2 |k(u)| < \infty$. $\int k = 1$.

$$\int (\mathbb{E}(\hat{p}(x)) - p(x))^2 dx = \frac{h^4}{4} \mu_2(K)^2 \int (p''(x))^2 dx + o(h^4).$$

pf) $\mathbb{E}(\hat{p}(x)) - p(x) = \int (K_h * p) - p(x_1)$

$$= \frac{1}{2} \mu_2(K) \cdot p''(x_1) h^2 + \dots = h^2 \cdot \int_0^1 \int_0^1 t^2 k(t) \{ p''(x - wht) - p''(x_1) \} \cdot (1-w) dw dt$$

$$\int (\mathbb{E}(\hat{p}(x)) - p(x))^2 dx = \frac{h^4}{4} \mu_2(K)^2 \int (p''(x))^2 dx +$$

$$+ \int \mu_2(K) \cdot p''(x_1) h^4 \cdot \int_0^1 \int_0^1 t^2 k(t) \{ p''(x - wht) - p''(x_1) \} \cdot (1-w) dw dt \cdot dx$$

$$+ h^4 \cdot \int_0^1 \int_0^1 \int_0^1 \int_0^1 t^2 k(t) k(s) \{ p''(x - wht) - p''(x_1) \} \{ p''(x - ws) - p''(x_1) \} \cdot (1-w_1)(1-w_2) dw_1 dw_2 dt ds \cdot dx$$

$h \rightarrow 0$, pointwise tending to zero.

Dominating term: $\left| \left(\int_0^1 \int_0^1 t^2 k(t) \cdot (1-w) \{ p''(x - wht) - p''(x_1) \} \cdot dw dt \right) \cdot \mu_2(K) \cdot p''(x_1) \right|$

$$\leq \mu_2(K) |p''(x_1)| \cdot \left\{ \int_{|H| \geq \frac{\sigma}{ah}} + \int_{|H| < \frac{\sigma}{ah}} \right\}$$

$$\leq \mu_2(K) |p''(x_1)| \left[\left(\sup_{|y| < \frac{\sigma}{ah}} |p''(x-y) - p''(x_1)| \right) \cdot \frac{1}{2} \int t^2 k(t) + \left(\max_{|t| \geq \frac{\sigma}{h}} |t|^3 k(t) \right) \cdot \frac{1}{\sigma} \int |p''(x)| dx \right]$$

$$+ \left(\int_{|H| \geq \frac{\sigma}{h}} |t|^2 k(t) dt \right) |p''(x_1)|$$

For appropriate $\sigma = \sigma(x)$, dominated ($\int p''^2 < \infty$)

Thm 2.3 p bounded $\int k^2 < \infty$, then

$$\int \text{var}(\hat{p}(x)) dx = \frac{1}{nh} \int k^2 + o\left(\frac{1}{nh}\right).$$

pf. ~~$\text{var}(\hat{p}(x)) = \frac{1}{nh} \int k^2(x) dx + o\left(\frac{1}{nh}\right)$~~

~~$\int k^2(x) dx = \int \int k^2(x) dx = \int \int k^2(x) dx = \int \int k^2(x) dx$~~

$$pf \quad var(\hat{p}(x)) = \frac{1}{nh} \int K^2(u) \hat{p}(x-hu) du - \frac{1}{n} \left(\int K(u) \hat{p}(x-hu) du \right)^2$$

$$\int var(\hat{p}(x)) dx = \frac{1}{nh} \int K^2(u) du - \frac{1}{n} \int \left(\int K(u) \hat{p}(x-hu) du \right)^2 dx.$$

$$\int \left(\int K(u) \hat{p}(x-hu) du \right)^2 dx \xrightarrow{h \rightarrow 0} \int p(x)^2 dx$$

Dominated term.

Optimal bandwidth: Minimizing MISE.

$$h_{opt} = \left[\frac{\int K^2}{(\mu_2(K))^2 \int (f'')^2} \right]^{\frac{1}{5}} n^{-\frac{1}{5}}.$$

$$\inf MISE = \frac{5}{4} \left\{ \mu_2(K) \left(\int K^2 \right)^{\frac{1}{2}} \right\}^{\frac{4}{5}} \left\{ \int (f'')^2 \right\}^{\frac{1}{5}} n^{-\frac{4}{5}} + o(n^{-\frac{4}{5}}).$$

$$\text{Minimize } \mu_2(K) \cdot \left(\int K^2 \right)^{\frac{1}{2}}.$$

→ Epanechnikov kernel.

$$K^*(u) = \frac{3}{4\sqrt{5}} \left(1 - \frac{u^2}{5} \right) I_{(-\sqrt{5}, \sqrt{5})}(u).$$

$$pf \quad L(K) = \int K^2(u) du + \lambda_1 \int u^2 K(u) du + \lambda_2 \int K(u) du.$$

$$\rightarrow 2K(u) + \lambda_1 u^2 + \lambda_2 = 0.$$

$$K^*(u) = \frac{3}{4\sqrt{5}} \left(1 - \frac{u^2}{5} \right) I_{(-\sqrt{5}, \sqrt{5})}(u).$$

Boundary Effect

$$p \text{ supported } [0, \infty). \quad \text{supp}(K) = [-1, 1]. \quad \mu_j(K; \alpha) = \int_{-1}^{\alpha} u^j K(u) du.$$

Theorem 2.4. $r = \alpha h$. $p^{(r)}$ right continuous at 0. $p^{(r)} \in L^1$.

$|u|^{r+1} K(u) \rightarrow 0$ as $|u| \rightarrow \infty$ and $\int |u|^r |K(u)| < \infty$. Then, $h \rightarrow 0$.

$$(K_h * p)(x) = \sum_{j=0}^r \frac{(-1)^j \mu_j(K; \alpha)}{j!} p^{(j)}(x) h^j + o(h^r)$$

$$pf \quad (K_h * p)(x) = \int_{-1}^x K(u) f(x-hu) du = \int_{-1}^x K(u) \left(\sum_{j=0}^r \frac{(-1)^j h^j}{j!} p^{(j)}(x) \right) + \frac{(-1)^r h^r}{(r-1)!} \left(\int_0^1 (f^{(r)}(x-wh) - f^{(r)}(x)) (1-w)^{r-1} dw \right) (h-u)^{r-1} du$$

$$= \sum_{j=0}^r \frac{(-1)^j \mu_j(K; \alpha)}{j!} p^{(j)}(x) h^j + \frac{(-1)^r h^r}{(r-1)!} \int_{-1}^x \int_0^1 (f^{(r)}(x-wh) - f^{(r)}(x)) (1-w)^{r-1} dw K(u) du$$

$$\text{Remainder; } \int_{-1}^x \int_0^1 K(u) \cdot (1-w)^{r-1} \left(f^{(r)}(x-wh) - f^{(r)}(x) \right) dw du \quad \text{and LDCT.}$$

$$\leq 2 \sup_{|y| \leq 4h} |f^{(r)}(y) - f^{(r)}(x)|$$

$$\rightarrow o(h^r).$$

$$(K_h \mp p)(x) - p(x) = (\mu_0(K; \alpha) - 1) p(x) - \mu_1(K; \alpha) p'(x) \cdot h + \frac{1}{2} \mu_2(K; \alpha) p''(x) h^2 + o(h^2).$$

$$= (\mu_0(K; \alpha) - 1) (p(0+) + \alpha h p'(0+) + \frac{1}{2} (\alpha h)^2 p''(0+) + o(h^2))$$

$$- \mu_1(K; \alpha) (p'(0+) + \alpha h p''(0+) + o(h)) p'(x) \cdot h$$

$$+ \frac{1}{2} \mu_2(K; \alpha) p''(0+) h^2 + o(h^2)$$

$$= (\mu_0(K; \alpha) - 1) p(0+) + (\alpha (\mu_0(K; \alpha) - 1) - \mu_1(K; \alpha)) p'(0+) \cdot h$$

$$+ \frac{1}{2} [\alpha^2 (\mu_0(K; \alpha) - 1) - 2\alpha \mu_1(K; \alpha) + \mu_2(K; \alpha)] p''(0+) \cdot h^2 + o(h^2).$$

$$\text{var}(\hat{p}(x)) = \frac{1}{n} \text{var}\left(\frac{\sum_{i=1}^n K_h(x_i - x)}{K_h(x - x_1)}\right) = \frac{1}{nh} \mu_0(K; \alpha) \cdot p(x) + o\left(\frac{1}{nh}\right).$$

Boundary Kernel

For bias & variance, we only need: $\int |K(u)| du \rightarrow 0$, $\int |K(u)|^2 du < \infty$.

$$\int |K(u)| du \rightarrow 0, \quad \int |K(u)|^2 du < \infty.$$

$\mu_0(K; \alpha) = 1$, $\mu_1(K; \alpha) = 0$, such kernel $K(\cdot; \alpha)$.

$$K(u; \alpha) = (\alpha u + \frac{1}{2} u^2) K(u).$$

$$\int_{-1}^1 \begin{pmatrix} 1 & u \\ u & u^2 \end{pmatrix} K(u) du \cdot \begin{pmatrix} \alpha \\ b(\alpha) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha \\ b(\alpha) \end{pmatrix} = \begin{pmatrix} \mu_0(K; \alpha) & \mu_1(K; \alpha) \\ \mu_1(K; \alpha) & \mu_2(K; \alpha) \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\therefore K(u; \alpha) = \frac{\mu_2(K; \alpha) - \mu_1(K; \alpha)u}{\mu_0(K; \alpha)\mu_2(K; \alpha) - \mu_1(K; \alpha)^2} \cdot K(u) \cdot \mathbb{1}_{(-1, 1)}(u).$$

Higher Order Kernel

K : order $l \Leftrightarrow \int K = 1$, $\mu_j(K) = 0$, $j = 1, \dots, l-1$, $\mu_l(K) \neq 0$, but finite.

Bias: $p^{(l)}$ continuous at x , $p^{(l)} \in L_1$, $\int |K(u)|^2 du \rightarrow 0$.

$$\text{then bias} = \frac{(-1)^l}{l!} \mu_l(K) \cdot p^{(l)}(x) h^2 + o(h^2)$$

$$\text{Variance: } \frac{1}{nh} \mu_0(K) p(x) + o\left(\frac{1}{nh}\right).$$

$$\text{Integral: } \int \text{bias}^2 = \frac{1}{2l!^2} \mu_l(K)^2 \int (p^{(l)}(x))^2 dx \cdot h^{2l} + o(h^{2l})$$

$$\int \text{var} = \frac{1}{nh} \mu_0(K) \int (p(x))^2 dx + o\left(\frac{1}{nh}\right)$$

MISE formula.

Construction of higher order kernels.

K_{2l} : symmetric l -th order kernel. (l : even).

$$\left(\int K = 1, \mu_j(K) = 0 \text{ for } j=1, \dots, l-1 \right)$$

$$\begin{cases} K_{2l+2}(x) = \frac{l+1}{l} K_{2l}(x) + \frac{1}{l} x K_{2l}'(x). \\ K_{2l+2}(x) = 2K_{2l}(x) - K_{2l}^* K_{2l}(x). \end{cases}$$

Local Bandwidth

bias: $\frac{1}{2} \mu_2(K)^2 h^2 \cdot p''(x) \rightarrow \text{bias}^2 = \frac{1}{4} \mu_2(K)^2 \cdot p''(x)^2 \cdot h^4.$

variance: $\frac{1}{nh} \mu_0(K) \cdot p(x).$

$$h_{opt}(x) = \left(\frac{p(x) \cdot \int K^2}{\mu_2(K)^2 p''(x)^2} \right)^{1/5} n^{-1/5}.$$

$$\text{MISE}(\hat{f}(\cdot; h(\phi))) = \int \frac{3}{4} \mu_2(K)^2 p''(x)^2 \cdot \left(\frac{p(x) \cdot \int K^2}{\mu_2(K)^2 p''(x)^2} \right)^{4/5} n^{-4/5} dx + o(n^{-4/5})$$

$$= \frac{3}{4} \int \mu_2(K)^{2/5} p''(x)^{2/5} \cdot p(x)^{4/5} \left(\int K^2 \right)^{4/5} n^{-4/5} dx + o(n^{-4/5})$$

$$= \frac{3}{4} n^{-4/5} \mu_2(K)^{2/5} \left(\int K^2 \right)^{4/5} \cdot \int p(x)^{4/5} p''(x)^{2/5} dx + o(n^{-4/5})$$

$$\inf_{h>0} \text{MISE}(\hat{f}(\cdot; h)) = \frac{3}{4} \left\{ \mu_2(K) \int K^2 \right\}^{2/5} \left(\int (f''(x))^2 \right)^{1/5} n^{-4/5} + o(n^{-4/5}).$$

$$\int p(x)^{4/5} p''(x)^{2/5} dx \leq \left(\int p''(x)^2 \right)^{1/5} \left(\int p(x)^{4/5} \right)^{4/5} \quad (\text{Holder}).$$

Variable Bandwidth

$$\hat{f}(x|h, a(\cdot)) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h a(x_i)} K\left(\frac{x-x_i}{h a(x_i)}\right).$$

$$\text{bias} = \frac{1}{h} \int \frac{p(u)}{a(u)} K\left(\frac{x-u}{h a(u)}\right) du - p(x).$$

$$\text{var} = \frac{1}{h} \int \frac{p(u)}{a(u)^2} K^2\left(\frac{x-u}{h a(u)}\right) du \cdot \frac{1}{nh} - \left\{ \frac{1}{h} \int \frac{p(u)}{a(u)} K\left(\frac{x-u}{h a(u)}\right) du \right\}^2 h^{-1}.$$

$$* J(x|h) = \frac{1}{h} \int p(z) L\left(\frac{x-z}{h a(z)}\right) dz.$$

with $\int (1+|z|^{k+1}) |L(z)| < \infty$ and $\frac{1}{a(\cdot)}, p(\cdot)$ function with l bounded, continuous derivative.

$\Rightarrow J(x|h)$ expansion coefficient of h^k is $\mu_k(L).$

$$\text{bias} = \frac{1}{2} \mu_2(K) (\alpha^2 f)''(\alpha) \cdot h^2 + \frac{1}{24} \mu_4(K) (\alpha^4 f)''''(\alpha) h^4 + o(h^4)$$

$$\text{var} = \mu_0(K) \frac{f(\alpha)}{\alpha f'(\alpha)} \cdot \frac{1}{nh} + o\left(\frac{1}{nh}\right)$$

$\alpha = p^{-1/2}$ min bias reduction.

Derivative Estimation

$$\begin{aligned} \hat{p}^{(r)}(x;h) &= \frac{1}{nh^{r+1}} \sum_{i=1}^n K^{(r)}\left(\frac{x-x_i}{h}\right) \\ &= \frac{1}{nh^r} \sum_{i=1}^n (K^{(r)})_h(x-x_i) \end{aligned}$$

$\left\{ \begin{array}{l} p^{(r+2)} \text{ continuous at } x, p^{(r+2)} \in L_1 \\ |t|^2 K(t) \rightarrow 0 \text{ as } |t| \rightarrow \infty \\ \int |t|^2 K(t) dt < \infty \text{ symmetric } \int K = 0 \end{array} \right.$

then bias = $\frac{1}{2} \mu_2(K) p^{(r+2)}(x) h^2 + o(h^2)$

$$\text{pf: } E(\hat{p}^{(r)}(x;h)) - p^{(r)}(x) = \frac{1}{h^r} \left\{ (K^{(r)})_h * p(x) \right\} - p^{(r)}(x)$$

$$\begin{aligned} \frac{1}{h^r} \left\{ (K^{(r)})_h * p(x) \right\} &= \frac{1}{h^r} \int_{-\infty}^{\infty} K^{(r)}\left(\frac{x-y}{h}\right) p(y) dy \\ &= \frac{1}{h^r} (-1)^r \int_{-\infty}^{\infty} K\left(\frac{x-y}{h}\right) p^{(r)}(y) dy = (-1)^r (K_h * p^{(r)})(x) \end{aligned}$$

$$\text{var} = \mu_0((K^{(r)})_h^2) \frac{f(x)}{\frac{1}{nh^{2r+1}}} \frac{1}{nh^{2r+1}}$$

$$h_{opt} \approx n^{-1/(2r+5)}, \text{ MSE} \sim n^{-4/(2r+5)} \quad (\text{bias}^2 \sim h^4, \text{var} \sim \frac{1}{nh^{2r+1}})$$

Distribution Function Estimation

$$\begin{aligned} \hat{F}(x;h) &= \int_{-\infty}^x \hat{f}(t;h) dt = \frac{1}{n} \sum_{i=1}^n \int_{-\infty}^x K_h(t-x_i) dt \\ &= \frac{1}{n} \sum_{i=1}^n L\left(\frac{x-x_i}{h}\right) \end{aligned}$$

$$L(t) = \int_{-\infty}^t K(u) du$$

~~Bias = $\frac{1}{2} \mu_2(K) p'(x) h^2 + o(h^2)$~~

$$\begin{aligned} E\left(\frac{1}{n} \sum_{i=1}^n L\left(\frac{x-x_i}{h}\right)\right) - \int_{-\infty}^x p(t) dt &= \int_{-\infty}^x L\left(\frac{x-x_i}{h}\right) p(x_i) dx_i - \int_{-\infty}^x p(t) dt \\ &= \int_{-\infty}^x \int_{-\infty}^{\frac{x-u}{h}} K(u) p(u) du du - \int_{-\infty}^x p(t) dt \\ &= \int_{-\infty}^{x-hu} p(u) du \cdot K(t) dt - \int_{-\infty}^x p(t) dt = \int_{-\infty}^x K(t) (F(x-hu) - F(x)) dt \\ &= (K_h * F)(x) - F(x) = \frac{1}{2} \mu_2(K) \cdot p'(x) \cdot h^2 + o(h^2) \end{aligned}$$

$$\begin{aligned}
\text{var} \{ \hat{F}_h \} &= \frac{1}{n} \text{var} \left\{ \int_{-\infty}^{\infty} K_h(t-x_i) dt \right\} \\
&= \frac{1}{n} \left[\mathbb{E} \left\{ \int_{-\infty}^{\infty} K_h(t-x_i) dt \right\}^2 - \left(\mathbb{E} \int_{-\infty}^{\infty} K_h(t-x_i) dt \right)^2 \right] \\
&= \frac{1}{n} \left[\int \left\{ \int_{-\infty}^{\infty} K_h(t-y) dt \right\}^2 f(y) dy - \{F(x)\}^2 + o(h) \right] \\
&= \frac{1}{n} \left[\int \left\{ \int_{-\infty}^{\frac{2y}{h}} K(u) du \right\}^2 f(y) dy - \{F(x)\}^2 + o(h) \right] \\
&= \frac{1}{n} \left[\int \int_{-\infty}^{\frac{2y}{h}} \int_{-\infty}^{\frac{2y}{h}} K(u_1) K(u_2) f(y) du_1 du_2 dy - \{F(x)\}^2 + o(h) \right] \\
&= \frac{1}{n} \left[\int \int_{-\infty}^{1-h(u_1 u_2)} K(u_1) K(u_2) \cdot f(y) dy du_1 du_2 - \{F(x)\}^2 + o(h) \right] \\
&= \frac{1}{n} \left[\int \int F(1-h(u_1 u_2)) K(u_1) K(u_2) du_1 du_2 - \{F(x)\}^2 + o(h) \right] \\
&= \frac{1}{n} \left[F(x) - h f(x) \cdot \int \int (u_1 u_2) \cdot K(u_1) K(u_2) \cdot du_1 du_2 - \{F(x)\}^2 + o(h) \right] \\
&= \frac{1}{n} F(x) (1 - F(x)) + \left(-\frac{2h}{n} f(x) \cdot \int_{-\infty}^{\infty} t \cdot K(t) L(t) dt \right) + o\left(\frac{h}{n}\right)
\end{aligned}$$

Asymptotic Normality

Theorem 2.5. (A5) p is continuous at x

(A6) $|u| K(u) \rightarrow 0$ as $|u| \rightarrow \infty$. $\int K^2 < \infty$.

If $h \rightarrow 0$, $nh \rightarrow \infty$ then

$$\frac{\hat{p}(x) - \mathbb{E}(\hat{p}(x))}{\sqrt{\text{var}(\hat{p}(x))}} \rightsquigarrow N(0, 1)$$

Corollary 2.3. (A5) p' continuous at x and $p' \in L^1$

(A4) K is symmetric. $|u|^3 K(u) \rightarrow 0$, $\int |u|^2 K(u) < \infty$. $\int K = 1$.

(A5) p is continuous at x

(A6) $|u| K(u) \rightarrow 0$ as $|u| \rightarrow \infty$, $\int K^2 < \infty$.

$$\hat{p}(x) - p(x) = \frac{1}{2} h^2 \mu_2(K) p''(x) + \frac{1}{\sqrt{nh}} \mu_0(K^2)^{1/2} p(x)^{1/2} Z_n + o_p\left(h^2 + \frac{1}{\sqrt{nh}}\right)$$

If $h \asymp n^{-1/5}$ then

$$n^{2/5} (\hat{p}(x) - p(x)) \rightsquigarrow N\left(\frac{1}{2} c^2 \mu_2(K) \cdot p''(x), c^{-1} \mu_0(K^2) p(x)\right)$$

pf Lindeberg-Feller thm.

$$\left\{ \begin{array}{l} \{Y_{ni}\}_{i=1}^n \text{ is row-wise independent.} \\ E Y_{ni} = 0, \text{ var } Y_{ni} < \infty, \quad \sigma_n^2 = \sum_{i=1}^n \text{var}(Y_{ni}). \\ \frac{1}{\sigma_n^2} \sum_{i=1}^n E(Y_{ni}^2 \mathbb{1}(|Y_{ni}| > \epsilon \sigma_n)) \rightarrow 0 \text{ as } n \rightarrow \infty, \quad \forall \epsilon > 0. \end{array} \right.$$

$$\Rightarrow \frac{1}{\sigma_n} \sum_{i=1}^n Y_{ni} \rightsquigarrow N(0,1).$$

$$Y_{ni} = (K_h(x - X_i) - K_h * p(x)) \quad ; \text{ row-wise i.i.d.}$$

$$\sigma_n^2 = \sum_{i=1}^n \text{var}(Y_{ni}) = n \text{var}(Y_{n1}) = n^2 \text{var}(\hat{p}(x)) = O\left(\frac{n}{h}\right).$$

$$\frac{1}{\sigma_n^2} \sum_{i=1}^n E(Y_{ni}^2 \mathbb{1}(|Y_{ni}| > \epsilon \sigma_n)) = \frac{1}{\sigma_n^2} n \cdot E(Y_{n1}^2 \mathbb{1}(|Y_{n1}| > \epsilon \sigma_n))$$

$$\stackrel{||^{\wedge}}{=} \mathbb{E} \left[\mathbb{1} \left\{ |K_h(x - X_1)| > \frac{\epsilon}{2} \sigma_n \right\} \cup \left\{ |K_h * p(x)| > \frac{\epsilon}{2} \sigma_n \right\} \right]$$

if p is continuous, $(K_h * p)(x) \rightarrow p(x)$ as $h \rightarrow 0$.

$$E(Y_{ni}^2 \mathbb{1}(|Y_{ni}| > \epsilon \sigma_n)) \leq 2 |K_h(x - X_i)|^2 \mathbb{1}(|K_h(x - X_i)| > \frac{\epsilon}{2} \sigma_n) + 2 |K_h * p(x)|^2$$

$$\frac{1}{\sigma_n^2} E(Y_{ni}^2 \mathbb{1}(|Y_{ni}| > \epsilon \sigma_n)) \leq 2 E \left(|K_h(x - X_i)|^2 \mathbb{1}(|K_h(x - X_i)| > \frac{\epsilon}{2} \sigma_n) \right) + \frac{n}{\sigma_n^2} + \frac{2n}{\sigma_n^2} |K_h * p(x)|^2$$

$$\sigma_n^2 = O\left(\frac{n}{h}\right) \hookrightarrow \leq C \cdot \int K^2(u) \mathbb{1}(|K(u)| > \frac{\epsilon}{2} h \sigma_n) f(x - hu) du + C_2 h$$

$$\rightarrow 0$$

□

Oversmoothing : Bandwidth Selection.

$$h_{opt} = \left[\frac{\int K^2}{(\mu_2(K))^2 \int p''^2} \right]^{1/5} n^{-1/5}.$$

$$\sup_{p \in \mathcal{P}(f, \sigma)} \left(\frac{\int K^2}{(\mu_2(K))^2 \int p''^2} \right) = \left(\frac{243 \int K^2}{35 \mu_2(K)^2} \right) \sigma^4.$$

$$\Rightarrow h_{os} = \left(\frac{243 \int K^2}{35 \mu_2(K)^2} \right)^{1/5} \hat{\sigma} \cdot n^{-1/5}.$$

Normal scale.

$$\text{For } p = f \hat{\sigma}.$$

$$\hat{h}_{NS} = \left(\frac{8\pi \int K^2}{3 \mu_2(K)^2} \right)^{1/5} n^{-1/5} \cdot \hat{\sigma}.$$

Choose $\hat{\sigma}$ as.

standard deviation
Quartile range

$$\frac{\hat{F}^{-1}(3/4) - \hat{F}^{-1}(1/4)}{\hat{F}^{-1}(3/2) - \hat{F}^{-1}(1/4)}$$

Least squares cross-validation, Bandwidth Selection.

$$\text{MISE} = \mathbb{E} \int (\hat{p}(x) - p(x))^2 dx.$$

$$\hat{p}_{-i}(x) = \frac{1}{n-1} \sum_{j \neq i} K_h(x - x_j).$$

$$\mathbb{E} \hat{p}_{-i}(x) = \mathbb{E} \int p(x) p(x) dx.$$

$$\therefore \text{MISE}(\hat{p}) = \int p(x)^2 dx = \mathbb{E} \left(\int \hat{p}(x)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{p}_{-i}(x_i) \right).$$

$$\text{LSCV}(h) := \int \hat{p}(x)^2 dx - \frac{2}{n} \sum_{i=1}^n \hat{p}_{-i}(x_i)$$

$$\text{if } K \text{ is symmetric } \Rightarrow \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n K_h(x_i - x_j) = \frac{2}{n(n-1)} \sum_{i < j} K_h(x_i - x_j)$$

$$\hat{h}_{\text{LSCV}} = \arg \min_{h > 0} \text{LSCV}(h)$$

$$\text{then } n^{1/5} \left(\frac{\hat{h}_{\text{LSCV}}}{h_{\text{opt}}} - 1 \right) \rightsquigarrow N(0, \sigma_{\text{LSCV}}^2)$$

K-fold

$A_1, \dots, A_K$

$$\hat{p}_{-A_k}(x) = \frac{1}{\sum_{i=1}^n \mathbb{1}(i \notin A_k)} \sum_{i \notin A_k} K_h(x - x_i).$$

$$\text{LSCV}^{K\text{-fold}}(h) = \int \hat{p}(x)^2 dx - \frac{2}{K} \sum_{k=1}^K \left(\sum_{i \in A_k} \hat{p}_{-A_k}(x_i) \right) / \sum_{i=1}^n \mathbb{1}(i \in A_k)$$

$$\hat{h}_{\text{LSCV}}^{K\text{-fold}} = \arg \min_h \text{LSCV}^{K\text{-fold}}(h).$$

Biased-Cross Validation

$$\text{MISE}(h) \approx \frac{1}{nh} \int K^2 + \frac{1}{4} h^4 \mu_2(K)^2 \int (f'')^2.$$

Estimate $\int (f'')^2$ by $\int (\hat{f}'')^2$. (Estimation of Derivative)

$$\mathbb{E} \int \hat{f}''^2 = \int f''^2 + \frac{1}{nh^5} \int (K'')^2 + O(h^3).$$

$$\text{BCV}(h) = \frac{1}{nh} \int K^2 + \frac{1}{4} h^4 \mu_2(K)^2 \cdot \left[\int \hat{f}''^2 - \frac{1}{nh^5} \int (K'')^2 \right].$$

Multivariate KDE

$X_1, \dots, X_n$ d-variate r.v. density $p: \mathbb{R}^d \rightarrow [0, \infty)$.

Kernels Product $K(x) = \prod_{j=1}^d K(x_j)$.

Radial $K(x) = K(\|x\|)$

Bandwidth - single $\hat{p}(x) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$

- Coord-wise $\hat{p}(x) = n^{-1} \left(\prod_{j=1}^d h_j\right)^{-1} \sum_{i=1}^n x\left(\frac{x_1 - x_{i1}}{h_1}, \dots, \frac{x_d - x_{id}}{h_d}\right)$.

- Matrix $\hat{p}(x) = n^{-1} \det(H)^{-1} \sum_{i=1}^n x(H^{-1}(x - x_i))$.

Theorem 3.1. Assume (A4) K is symmetric, $\lim_{|u| \rightarrow \infty} K(u) = 0$, $\int |u|^2 K(u) < \infty$, $\int K = 1$.

p is twice continuously differentiable at x . $\frac{\partial^2}{\partial x_j \partial x_k} p(x) \in L^1$.

$$\mathbb{E}(\hat{p}(x)) - p(x) = \frac{1}{2} \mu_2(K) \sum_{j=1}^d h_j^2 \frac{\partial^2}{\partial x_j^2} p(x) + o(h_1^2 + \dots + h_d^2).$$

Theorem 3.2. Assume (A6). $\lim_{|u| \rightarrow \infty} K(u) = 0$, $\int K^2 < \infty$, p continuous at x .

$$\text{var}(\hat{p}(x)) = n^{-1} h_1^{-1} \dots h_d^{-1} \mu_0(K^2) p(x) + o(n^{-1} h_1^{-1} \dots h_d^{-1}).$$

$$g(y) = g(x) + \sum_{|I|=1} \frac{1}{|I|!} g_{|I|}(x) (y-x)^I + o(\|y-x\|^r).$$

$$= g(x) + \sum_{|I|=1} \frac{1}{|I|!} g_{|I|}(x) (y-x)^I + \frac{1}{(r-1)!} \int_0^1 \sum_{|I|=r} \{g_{|I|}(x+u(y-x)) - g_{|I|}(x)\} (y-x)^I (1-u)^{r-1} du.$$

↳ Taylor Formula.

pf of Thm 3.1. & 3.2.

Lemma: Assume $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is r times partially continuously differentiable on nhbd of x .
 r th partial derivative of f is integrable.
 $\lim_{|t| \rightarrow \infty} |t|^{r+1} K(t) = 0$ as $|t| \rightarrow \infty$
 $\int |t|^r K(t) < \infty$.

$$\text{then } \mathbb{E} \left[\det(H)^{-1/2} K(H^{-1/2}(x - X_1)) \right] = f(x) + \sum_{|I|=1} \frac{(-1)^{|I|}}{|I|!} f_{|I|}(x) \int (H^{1/2} t)^I K(t) dt + o(\text{tr}(H^T H^{1/2})).$$

$$\text{pf of Thm 3.1. } |I|=1, \quad (-1) \sum_{i=1}^d \frac{\partial}{\partial x_i} f(x) \int (H^{1/2} t)_i K(t) dt$$

$$= (-1) (Df(x))^T H^{1/2} \int t K(t) dt = 0.$$

$$|I|=2: \text{ sum} = \frac{1}{2} \int (H^{1/2} t)^T D^2 f(x) (H^{1/2} t) K(t) dt$$

$$= \frac{1}{2} \int \text{tr} \{ H^{1/2} D^2 f(x) H^{1/2} \} t^T t K(t) dt = \frac{1}{2} \text{tr} \{ H^{1/2} D^2 f(x) H^{1/2} \} \mu_2(K) I$$

$$= \frac{1}{2} \mu_2(K) \cdot \text{tr}(D^2 f(x) H).$$

$$\text{var}(\hat{f}(x; H)) = n^{-1} \text{Det}(H)^{-1} \underbrace{\text{var} \left\{ \int K(H^{1/2}(x-x_1)) f(x_1) dx_1 \right\}}_{=}$$

$$\text{Det}(H)^{1/2} \int K^2(t) f(x-H^{1/2}t) dt = \text{Det}(H) \left\{ \int K(t) f(x-H^{1/2}t) dt \right\}^2,$$

$$\text{Det}(H)^{1/2} \int K^2(t) dt f(x) (1+o(1)).$$

$$= n^{-1} \text{Det}(H)^{-1/2} \int K^2(t) dt f(x) + o(n^{-1} \text{Det}(H)^{-1/2}).$$

MISE

Theorem 3.3. $p \in L^2$, twice partially continuously differentiable on $\mathbb{R}^d$. $\frac{\partial^2}{\partial x_j \partial x_k} p \in L^2$.

K symmetric, $\int u^2 K(u) du < \infty$, $\int K^2 < \infty$, $\int K = 1$.

$$\text{MISE}(h_1, \dots, h_d) = \frac{1}{4} (\mu_2(K))^2 \int \left(\sum_{j=1}^d h_j^2 \frac{\partial^2}{\partial x_j^2} p(x) \right)^2 dx + \frac{1}{n h_1 \dots h_d} \mu_0(K^2) + o(h_1^2 + \dots + h_d^2 + n^{-1} h_1^{-1} \dots h_d^{-1}).$$

$h_j = c_j \cdot n^{-1/(d+4)}$, then MISE minimized.

$$\text{LSCV}(h_1, \dots, h_d) = \int \hat{p}(x)^2 dx - 2n^{-1} \sum_{i=1}^n \hat{p}_i(x_i).$$

• Bandwidth Shrinkage Scheme

① choose marginal $\hat{h}_j^{\text{marg}}$ ② $\hat{\alpha} = \argmin \text{LSCV}(\alpha \hat{h}_1^{\text{marg}}, \dots, \alpha \hat{h}_d^{\text{marg}})$ ③ $\hat{h}_j = \hat{\alpha} \cdot \hat{h}_j^{\text{marg}}$

• Coordinate-wise bandwidth selection.

① initial $\hat{h}_j^{(0)}$ ② coordinate wise update.